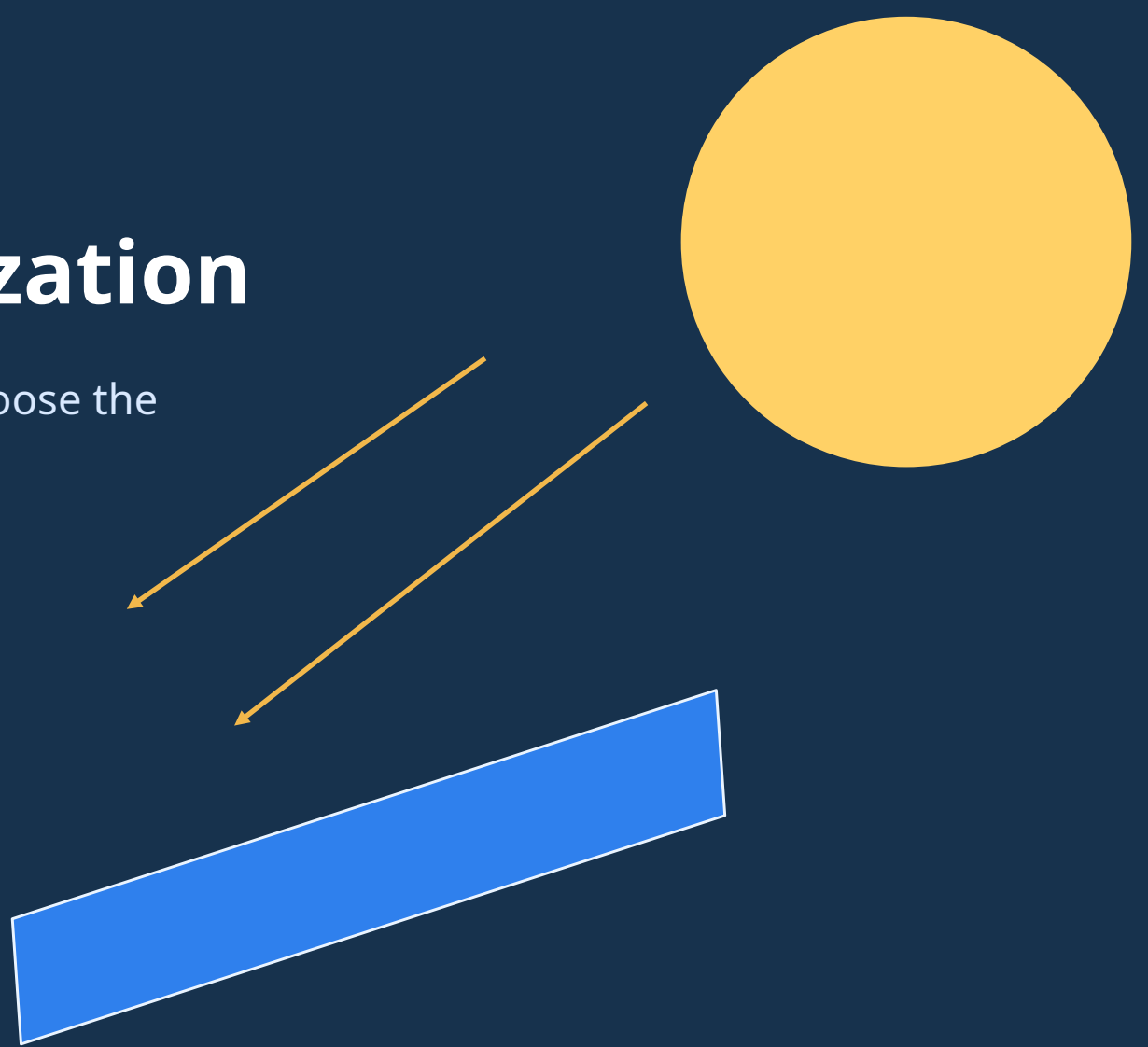


Solar Panel Tilt Optimization

A step-by-step presentation explaining how to choose the best fixed panel angle for maximum solar energy absorption.

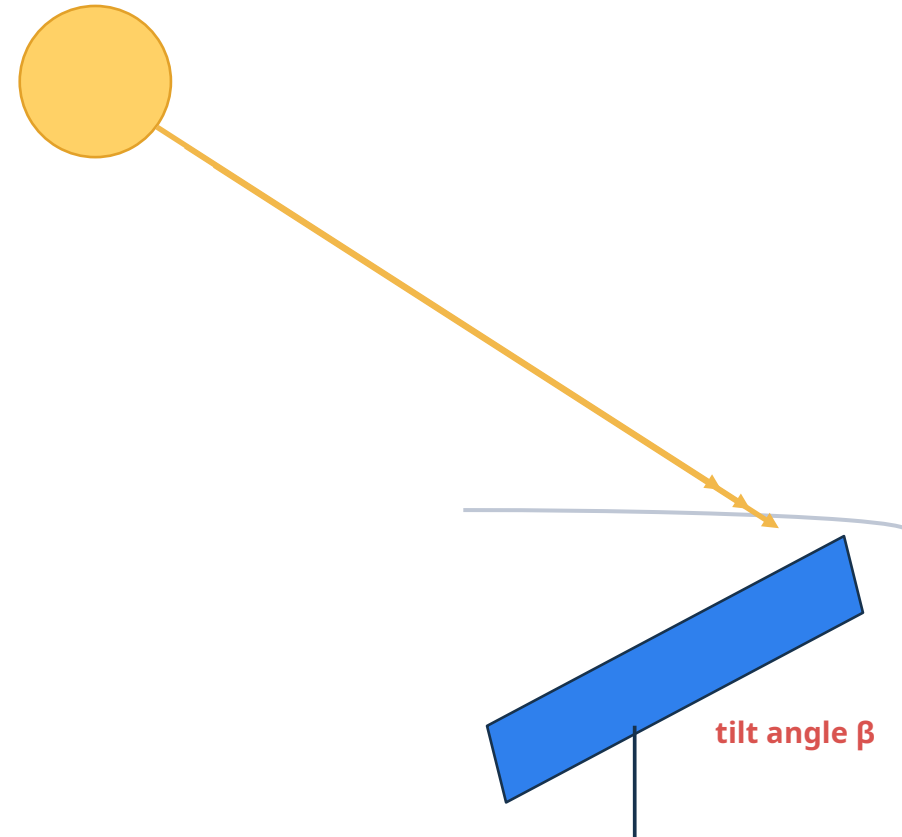
Case study: fixed, south-facing panel in Phoenix, Arizona



Presentation roadmap

What this presentation will teach, in order

What problem are we solving?
What choice are we allowed to make?
How do we turn that choice into math?
What limits keep the answer realistic?
How did the computer search for the best angle?
What result did we get, and what does it mean?



Main idea: this is not about building the panel. It is about choosing the best angle before the panel is installed.

1. Problem motivation and context

Why tilt angle matters in the real world

Solar panels make more electricity when more sunlight lands on the panel surface.

The Sun moves across the sky every day, and its height changes over the year.

A panel that is too flat may miss low winter Sun. A panel that is too steep may miss high summer Sun.

The installer usually chooses one fixed angle, so the decision matters for a long time.

Real users: homeowners, solar installers, businesses, utilities, and engineers.

Real decision: choose one fixed tilt angle that gives strong yearly energy absorption.

Real challenge: the “best” angle changes hour by hour, so we need a year-level optimization.

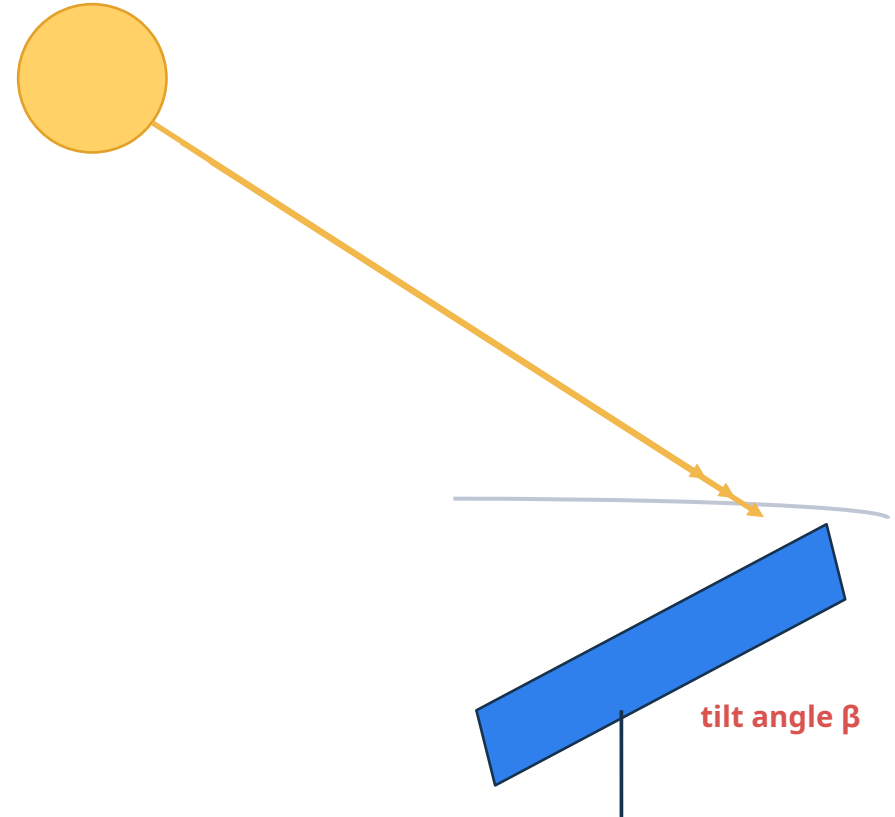
What exactly are we optimizing?

Keep the project well-scoped

We are choosing ONE number:

β = fixed panel tilt angle

The panel faces south the whole time.
The panel does not track the Sun.
We want the most solar energy over a full year.
The answer is measured in degrees from horizontal.



Example: $\beta = 0^\circ$ means the panel is flat. $\beta = 90^\circ$ means the panel is vertical like a wall.

2. Decision variable

The variable is the choice the optimizer controls

Decision variable

β = panel tilt angle

Units: degrees

Type: continuous variable

Bounds: $0^\circ \leq \beta \leq 90^\circ$

Continuous means β can be 31° , 31.7° , 31.72° , or any value in the allowed range.

Not binary. Not integer. The model does not force the angle to be a whole number.

The optimizer only changes β . Everything else is treated as input data.

Inputs are not decisions

Inputs describe the location, Sun, and weather

Latitude φ	where the system is on Earth
Solar zenith θ_z	how high or low the Sun is
Solar azimuth γ_s	which compass direction the Sun is in
DNI, DHI, GHI	direct, diffuse, and horizontal sunlight
Albedo ρ_g	how much light the ground reflects

Important distinction

A decision variable is what we choose.
An input is information the model uses.
The model uses Sun and irradiance data to judge whether a tilt angle is good.

In this presentation, Phoenix latitude is used for the demonstration case.

3. Objective function

The goal is written as a math expression

Plain English goal:

Choose β so the panel receives the most total solar energy over the year.

Math version:

Maximize $E(\beta) = \sum \text{over all hours} [\text{sunlight hitting the tilted panel}] \times \text{time step}$

This is better than saying “maximize sunlight” because it defines exactly what sunlight means in the model.

Objective function: the three sunlight pieces

Total panel energy has three parts

1. Direct beam

Sunlight that comes straight from the Sun. This depends strongly on the angle.

2. Diffuse sky

Light scattered by the atmosphere. It comes from the sky dome, not just one direction.

3. Ground reflection

Light that bounces off the ground and reaches the panel.

Full simplified expression:

$$E(\beta) = \sum t [DNI_t \max(0, \cos \theta_{i,t}(\beta)) + DHI_t(1+\cos\beta)/2 + \rho_g GHI_t(1-\cos\beta)/2] \Delta t$$

The summation symbol Σ means “add this up for every hour in the year.”

Angle of incidence: why tilt changes direct sunlight

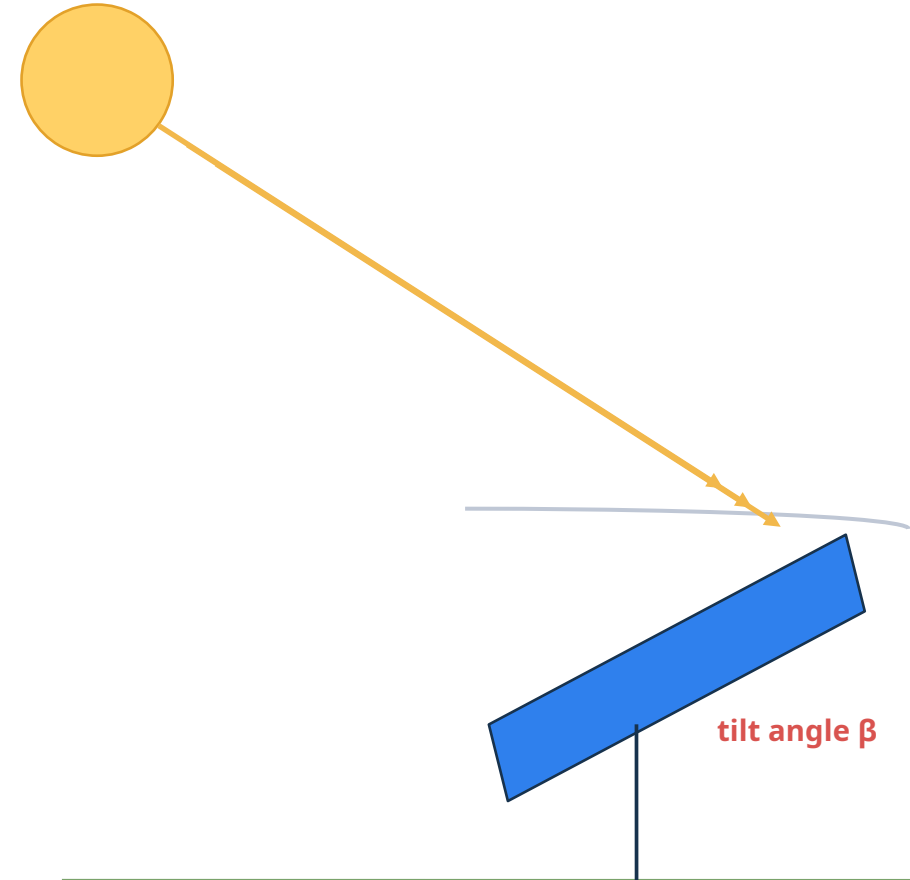
This is the key nonlinear part of the model

θ_i is the angle between:

the Sun ray coming toward the panel
the imaginary line sticking straight out of the
panel surface

If θ_i is small, sunlight hits the panel more directly.

If θ_i is large, sunlight hits at a shallow angle and less energy is captured.



Because $\cos(\theta_i)$ appears in the objective, the problem is nonlinear. Changing β changes cosine values, not just straight-line terms.

4. Constraints

Constraints keep the answer realistic

Main bound constraint
 $0^\circ \leq \beta \leq 90^\circ$

Meaning

The panel cannot tilt below the ground or past vertical in this simple fixed-mount model.

Sunlit direct-beam constraint
 $B_t(\beta) = \text{DNI}_t \max(0, \cos \theta_{i,t}(\beta))$

Meaning

If the Sun is behind the panel, direct beam contribution cannot be negative. It becomes zero.

Optional practical limit
 $\beta_{\min} \leq \beta \leq \beta_{\max}$

Meaning

A real mounting rack might only allow certain angles, like 10° to 60° .

5. Problem classification

What kind of optimization problem is this?

Classification:

Bounded nonlinear optimization

Bounded: β has lower and upper limits.

Nonlinear: the objective uses sine, cosine, and max functions.

Continuous: β can take decimal values.

One-dimensional: there is only one decision variable.

Why it is not LP

A linear program only has straight-line terms. This model has trigonometric functions, so it is not linear.

6. Solution methodology

How the computer finds the best tilt

Grid search in simple words:

Try $\beta = 0.0^\circ$, then 0.1° , then 0.2° , all the way to 90.0° .
For each β , calculate total yearly panel energy $E(\beta)$.
Keep track of the tilt with the highest $E(\beta)$.
Return that tilt as the best modeled answer.

Why grid search works well here

There is only one decision variable, so trying many possible angles is easy, transparent, and reproducible.

Step size used in the demonstration: 0.1° .
Smaller step sizes can make the result more precise.

Code and reproducibility

How someone else can repeat the work

Reproducible means another person can run the same steps and check the answer.

Inputs

Latitude, panel direction, hourly
Sun position, irradiance values,
albedo, and time step.

Loop

For each possible tilt angle,
compute the solar energy hitting
the tilted surface.

Output

Best tilt angle, best annual
energy, comparison to horizontal
panel, and a plot.

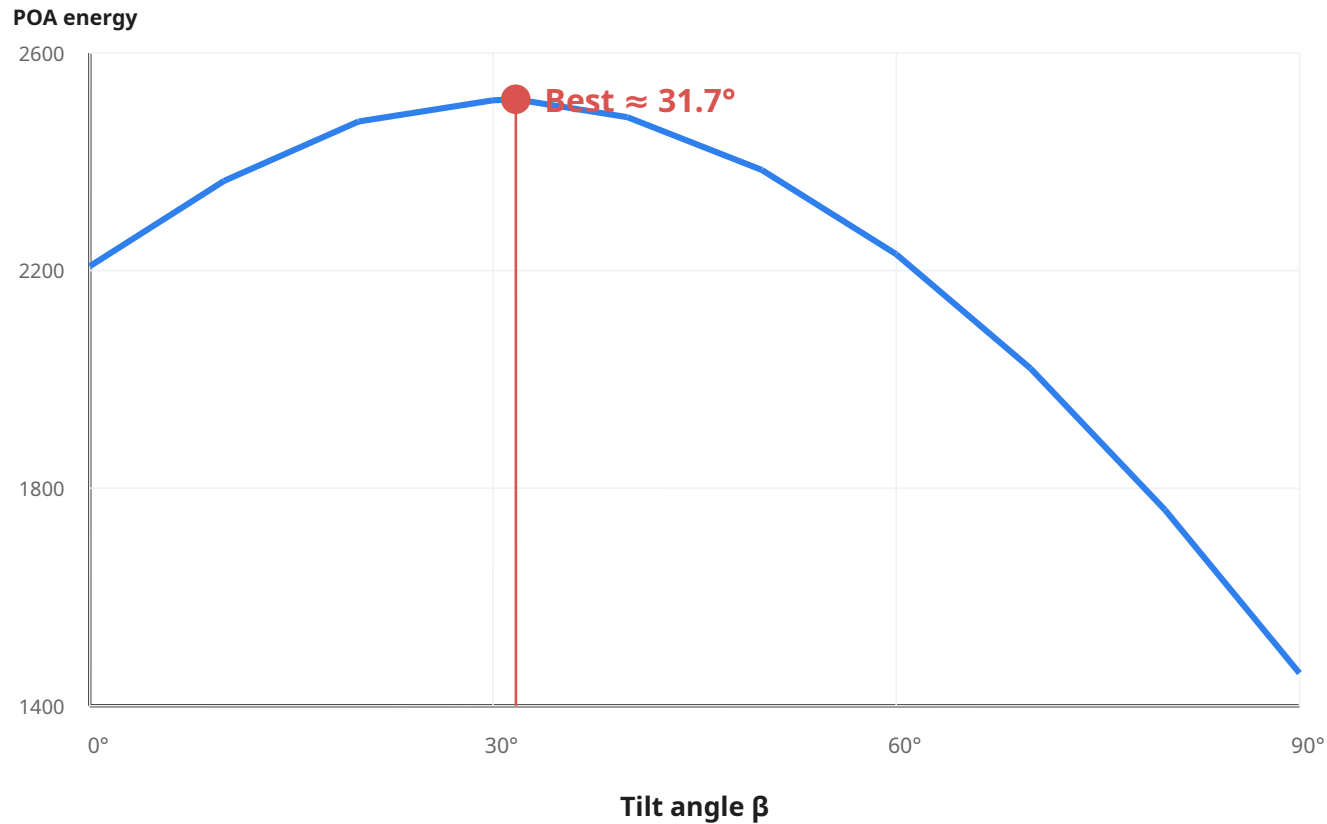
Tiny pseudocode:

```
best = none
for  $\beta$  from 0° to 90°:
    energy = calculate_E( $\beta$ )
    if energy is highest so far:
        best =  $\beta$ 
print(best)
```

The full report includes Python code so the result can be recreated instead of only described.

7. Results

The demonstration model found a best tilt near 31.7°



Best modeled tilt
 $\beta^* \approx 31.7^\circ$

Best modeled annual POA energy
 $\approx 2,514.8 \text{ kWh/m}^2$

Improvement vs horizontal panel
 $\approx 13.9\%$

The curve is smooth and fairly flat around the top, so nearby angles also perform well.

How to read the graph

The highest point is the best modeled answer

The horizontal axis is the tilt angle β .

The vertical axis is the modeled yearly solar energy hitting the panel.

The curve rises when the panel becomes better aligned with the Sun over the year.

After the best point, the curve drops because the panel becomes too steep for many hours.

Key takeaway

A tilt around 32° is the best result in this demonstration, but 30° to 33° is basically a very strong practical range.

This matters in real installation because mounting hardware may use standard angle settings instead of exact decimals.

What the answer means in plain language

Do not overthink the decimal

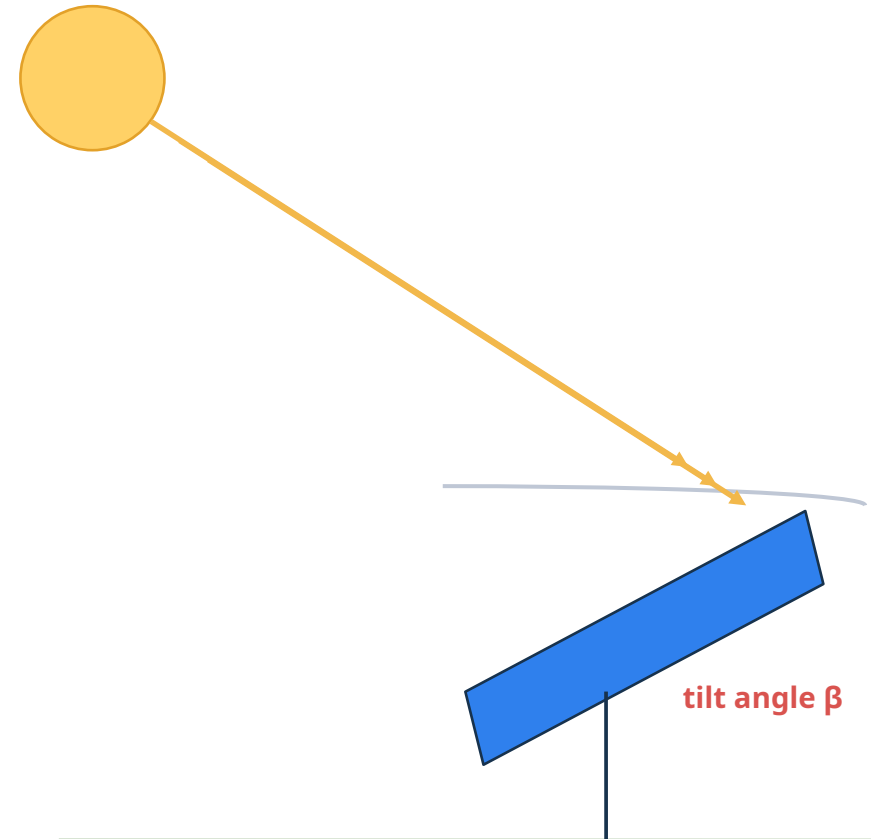
$$\beta^* \approx 31.7^\circ$$

This is the model's best fixed angle for the demonstration case.

It does not mean 31.7° is magically perfect in every real-world setting.

It means that, using the stated assumptions and input data, 31.7° gave the largest yearly total.

A practical installer might choose 30° , 32° , or another nearby rack setting if it is easier to build.



Optimization gives a mathematically best answer, but engineering still considers hardware, cost, roof slope, safety, and maintenance.

Assumptions and limits

What the model does not fully capture

The panel is fixed and south-facing.

The model uses an illustrative irradiance profile for demonstration.

Shading from trees, buildings, or other panel rows is not included.

Temperature losses, inverter losses, dirt, snow, and maintenance are not included.

The result should be updated with measured weather data for a final engineering design.

Why assumptions are okay

They make the first optimization clear enough to understand, solve, and explain. More details can be added later.

Future upgrade

Use real TMY or measured hourly weather data.

Final conclusion

The project turns a real solar design choice into an optimization problem

Decision variable: β , the fixed panel tilt angle in degrees.
Objective: maximize yearly plane-of-array solar energy.
Constraints: keep β physically realistic and prevent negative direct sunlight.
Classification: bounded nonlinear optimization.
Result: best demonstration tilt is about 31.7° for the Phoenix case.

Big picture

The best angle is found by testing possible tilts, calculating energy for each one, and choosing the angle with the highest total.

Sources

References used for the presentation and report

National Renewable Energy Laboratory (NREL). Solar Position Algorithm / PV modeling resources. <https://www.nrel.gov/>

Sandia National Laboratories PV Performance Modeling Collaborative. Plane-of-array irradiance and angle-of-incidence modeling. <https://pvpmc.sandia.gov/>

Duffie, J. A., & Beckman, W. A. Solar Engineering of Thermal Processes. Wiley.

OpenAI. ChatGPT. Used as a writing and organization aid for this project presentation, based on the user-provided project topic and generated report. September 3, 2026.

Instructor-provided project prompt/rubric. Optimization formulation requirements and bonus categories.

Note: ChatGPT was cited as a development aid. The scientific modeling concepts should be supported by engineering references such as NREL, Sandia, and solar engineering textbooks.